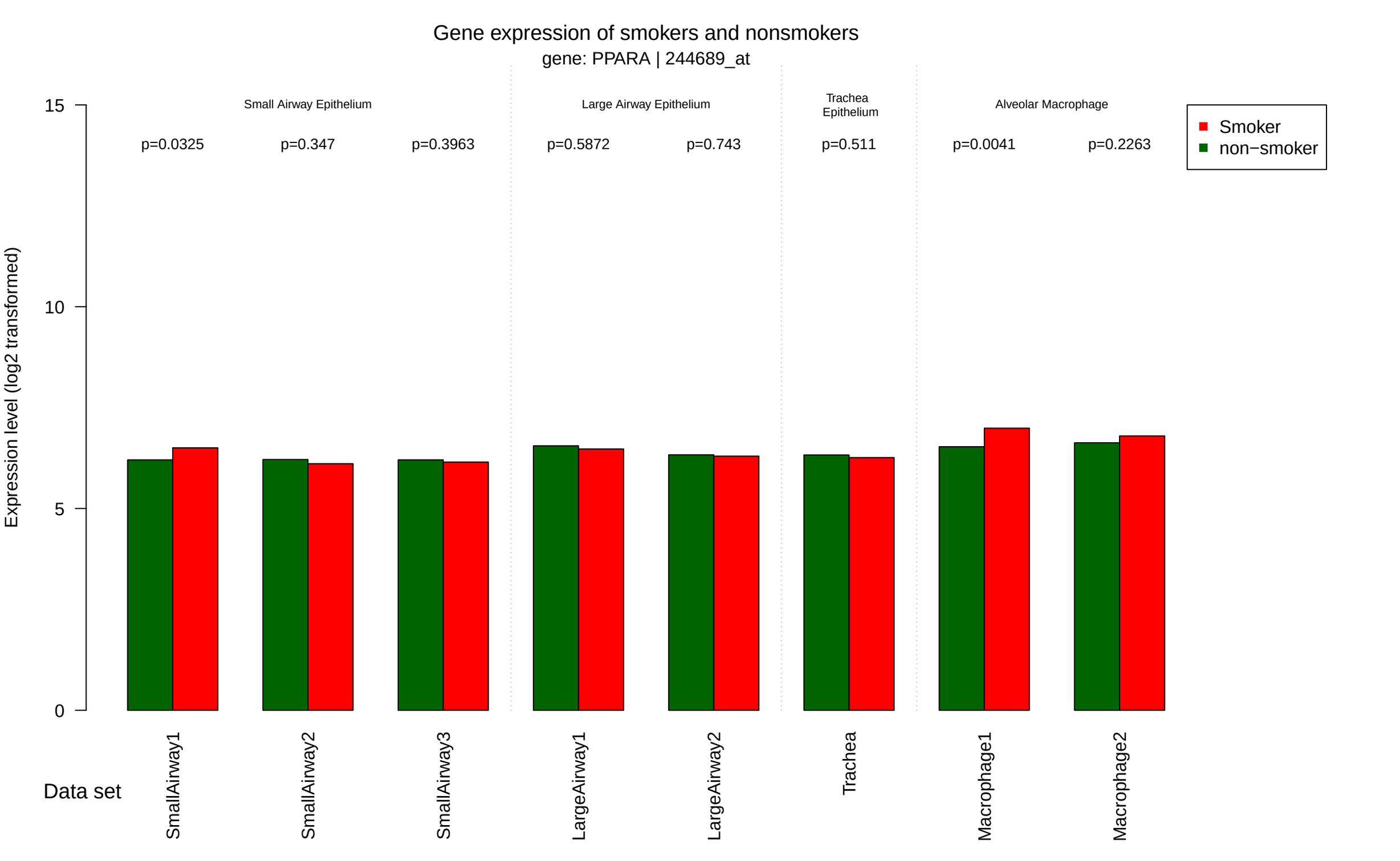
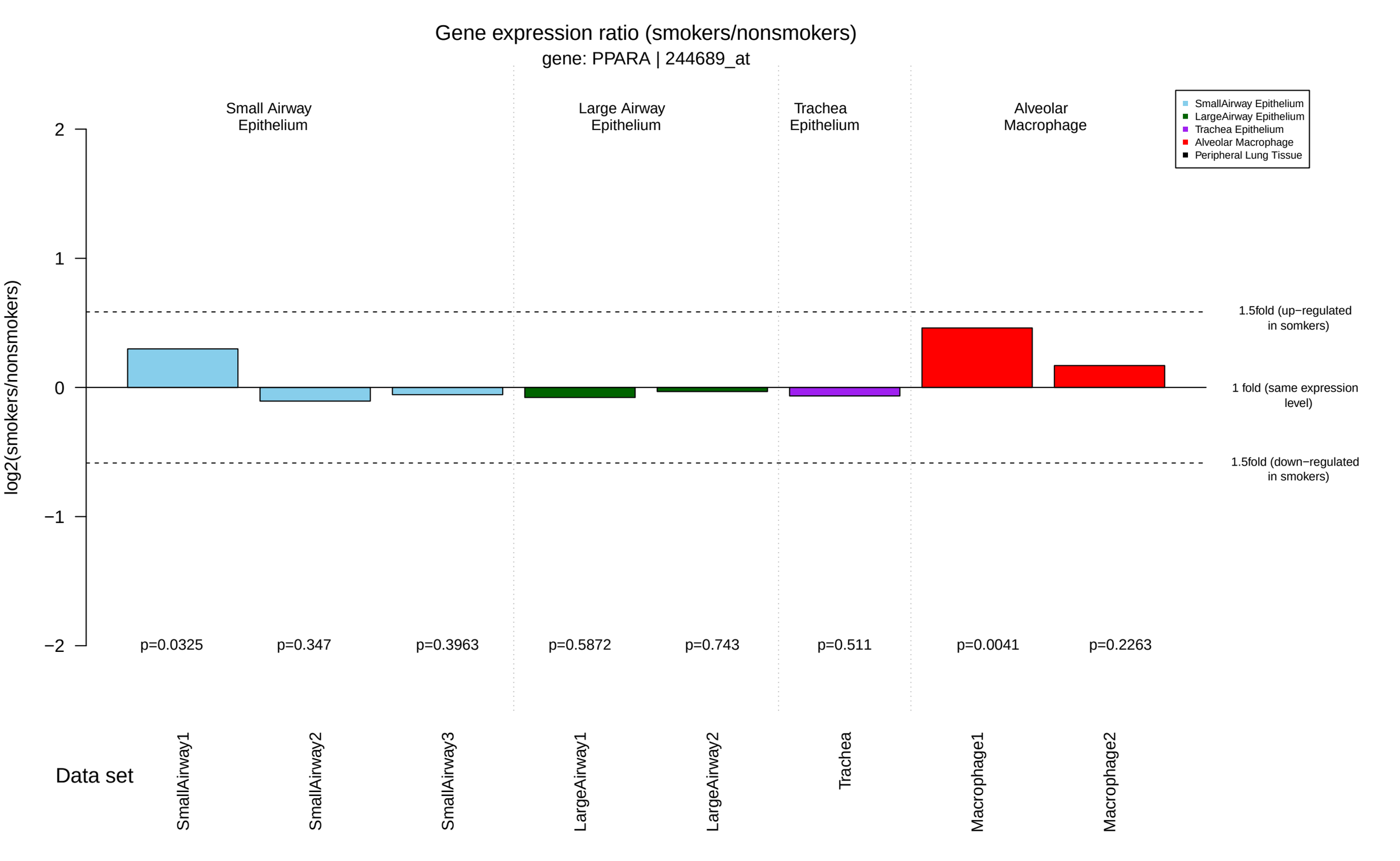
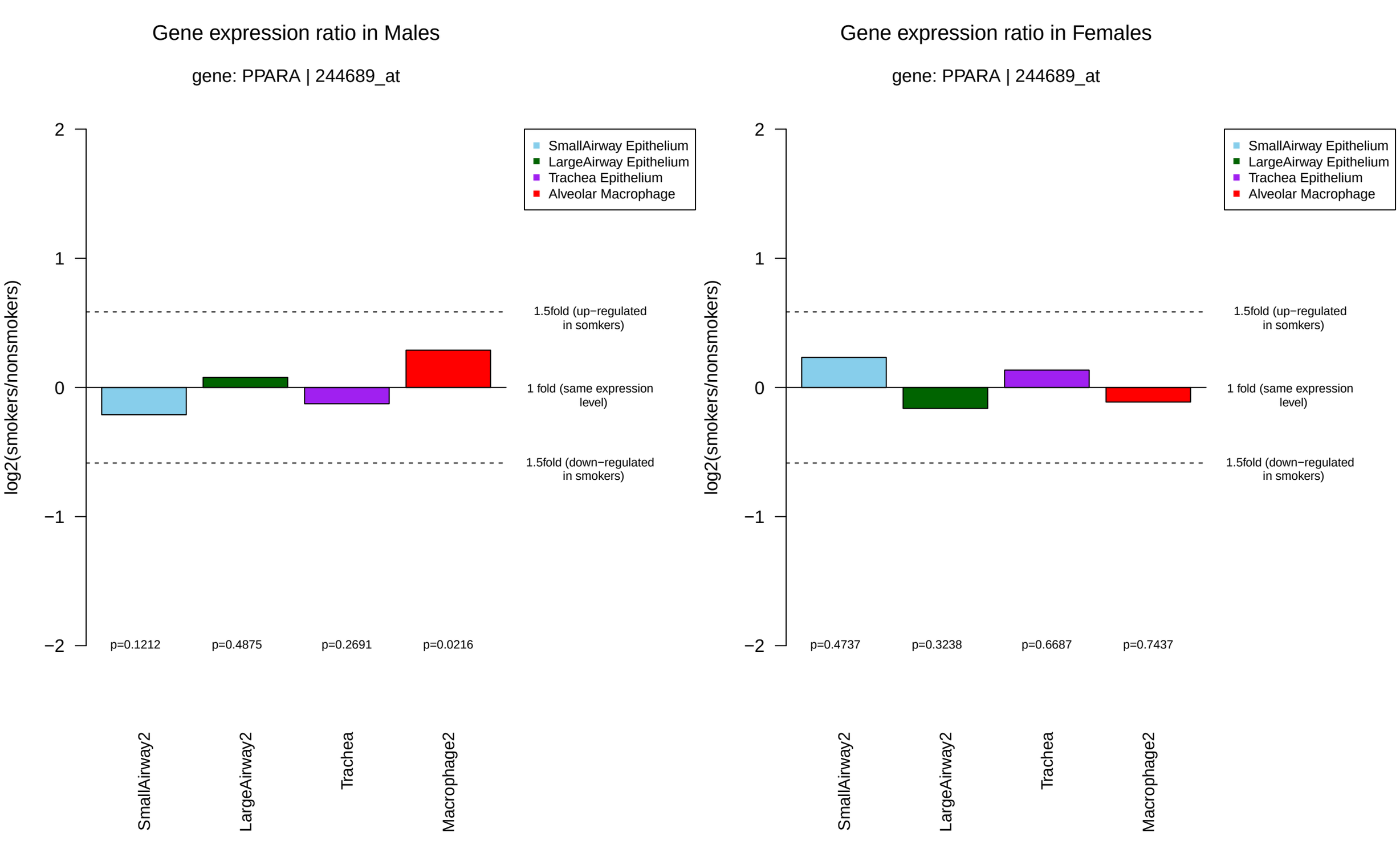
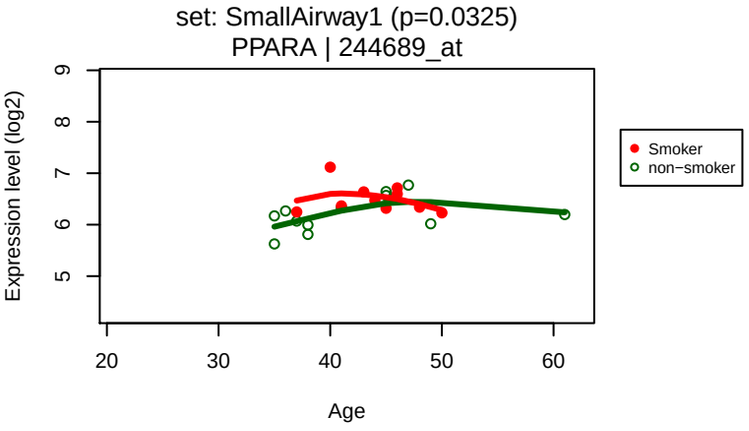
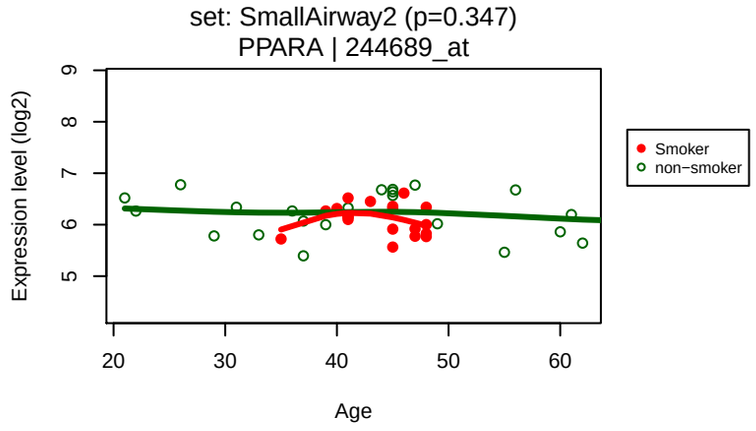
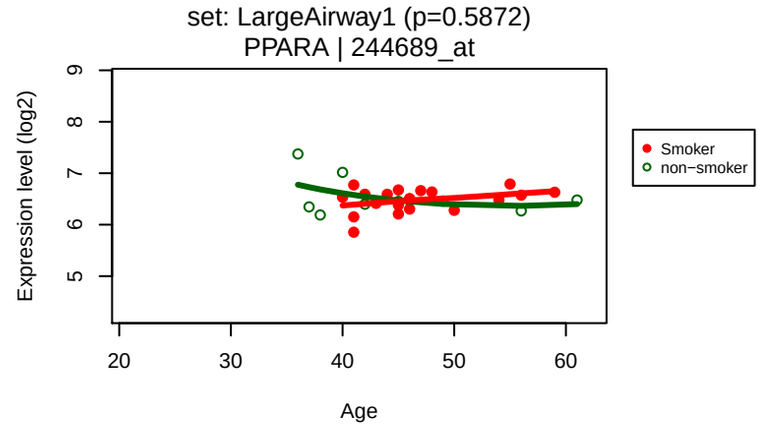

Gene expression of smokers and nonsmokers
gene: PPARA | 244689_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0325
p=0.347
p=0.3963
p=0.5872
p=0.743
p=0.511
p=0.0041
p=0.2263
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

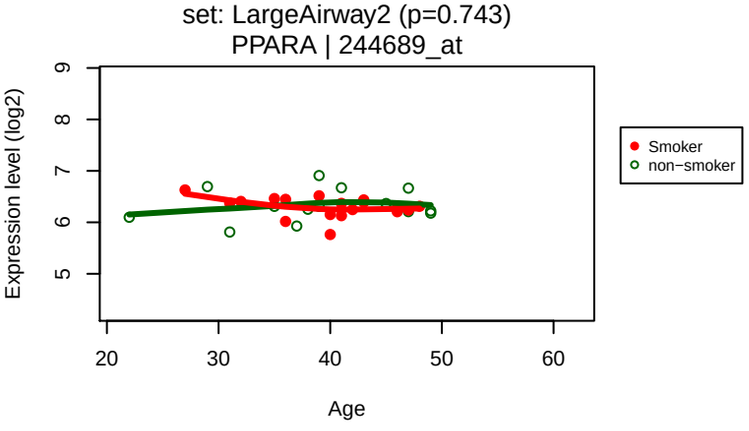

Gene expression ratio (smokers/nonsmokers)
gene: PPARA | 244689_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0325
p=0.347
p=0.3963
p=0.5872
p=0.743
p=0.511
p=0.0041
p=0.2263
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

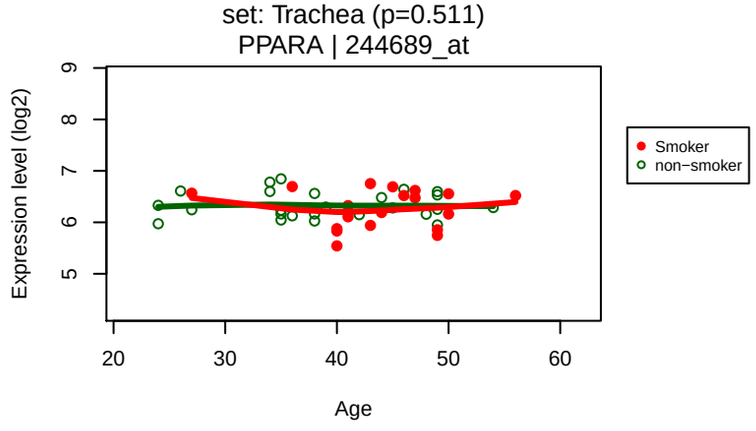

Gene expression ratio in Males
gene: PPARA | 244689_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1212
p=0.4875
p=0.2691
p=0.0216
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPARA | 244689_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4737
p=0.3238
p=0.6687
p=0.7437
SmallAirway2
LargeAirway2
Trachea
Macrophage2

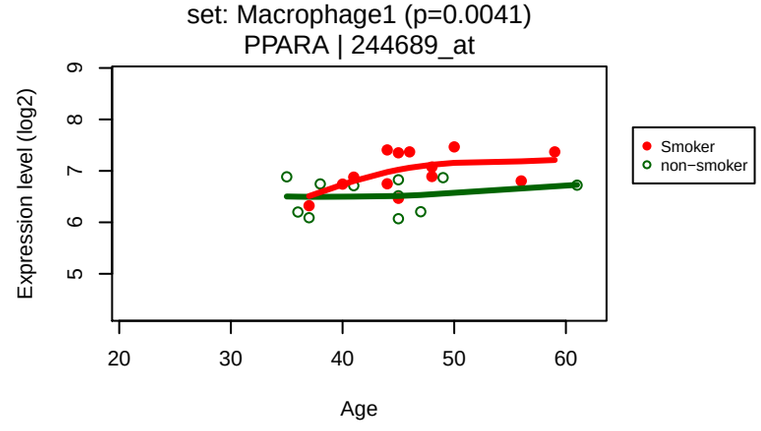

set: SmallAirway1 (p=0.0325)
PPARA | 244689_at
Expression level (log2)
Age
Smoker
non−smoker

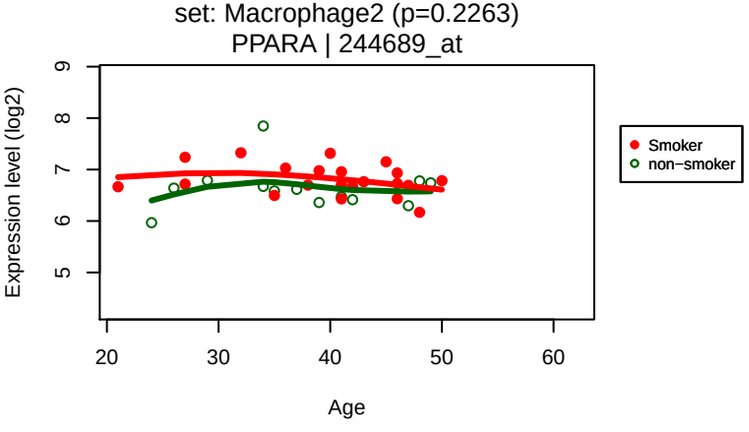

set: SmallAirway2 (p=0.347)
PPARA | 244689_at
Expression level (log2)
Age
Smoker
non−smoker

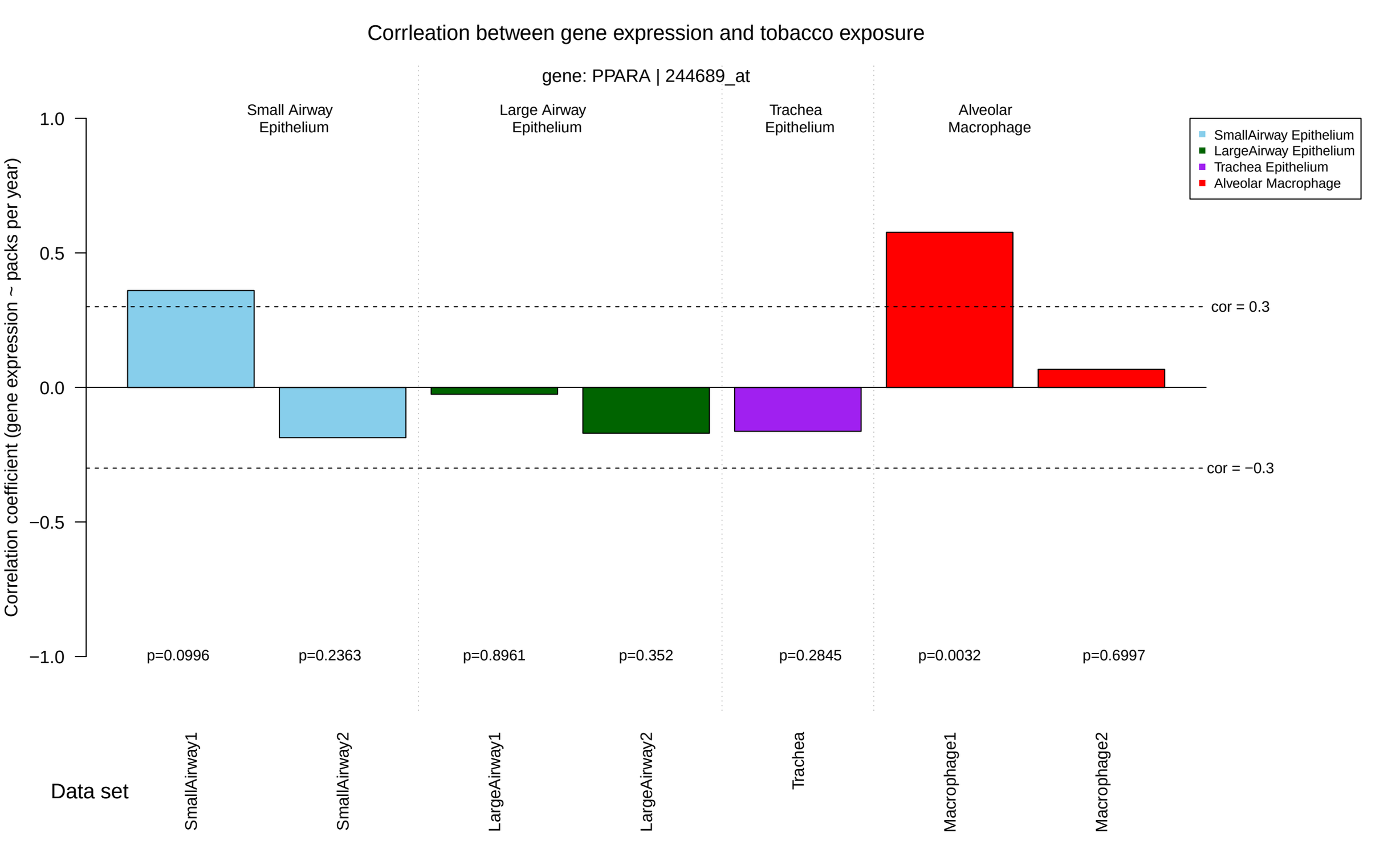

set: LargeAirway1 (p=0.5872)
PPARA | 244689_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.743)
PPARA | 244689_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.511)
PPARA | 244689_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0041)
PPARA | 244689_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2263)
PPARA | 244689_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPARA | 244689_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0996
p=0.2363
p=0.8961
p=0.352
p=0.2845
p=0.0032
p=0.6997
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2